

MAXIMUM LIKELIHOOD ESTIMATOR

16th August 2026

WHAT IS THE MOST LIKELY ANSWER?

The Maximum Likelihood Estimator, MLE, is possibly the most prevalent point estimator in statistics. It is unique in most cases and is an asymptotically unbiased estimator. The MLE is calculated by a method of estimation which is based on the likelihood of an obtained sample. We might ask, what does likelihood mean in a mathematical context? Let's denote the PMF at x by $p(x | \theta)$, if the model is discrete, and the PDF by $f(x | \theta)$, if the model is continuous. The likelihood of an IID sample $X_1, X_2, \dots, X_n$ is defined as

$$l(\theta | X_1, X_2, \dots, X_n) = \begin{cases} \prod_{i=1}^n p(X_i | \theta) & \text{discrete case} \\ \prod_{i=1}^n f(X_i | \theta) & \text{continuous case} \end{cases}. \text{ In words, the likelihood is a measure of how well a}$$

set of data supports a particular model or parameter values. Do note that to simplify notations, we usually drop the $X_1, X_2, \dots, X_n$ part from the notation for likelihood and write it simply as $l(\theta)$ when there is no scope of confusion.

The Maximum Likelihood Estimate of θ is the value of θ , $\hat{\theta}$, that maximises the likelihood, *i.e.*, makes the data most probable or most likely. The MLE is consistent for the true parameter when certain conditions are satisfied. Rather than maximising the likelihood itself, it is often easier to maximise its logarithm, which is an equivalent problem because logarithm is a strictly increasing function. The log-likelihood is $L(\theta) = \ln(l(\theta)) = \sum_{i=1}^n \ln f(X_i | \theta)$. If $L(\theta)$ is twice differentiable, the MLE, $\hat{\theta}$, satisfies $\frac{dL}{d\theta} \Big|_{\theta=\hat{\theta}} = 0$ and $\frac{d^2L}{d\theta^2} \Big|_{\theta=\hat{\theta}} < 0$.

AN EXAMPLE

What is the MLE for the Poisson distribution?

Solution: Let $X_1, X_2, \dots, X_n$ be n IID Poisson(λ). Assume that at least one of them is positive. The PMF of a Poisson distribution $p(x | \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$. By definition, the likelihood is $l(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$. Thus, the log-likelihood is $L(\lambda) = \ln \left(\prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \right) = \sum_{i=1}^n \ln \left(\frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \right) = \sum_{i=1}^n (\ln(e^{-\lambda}) + \ln(\lambda^{x_i}) - \ln(x_i!)) = \sum_{i=1}^n (x_i \ln(\lambda) - \lambda - \ln(x_i!)) = -n\lambda + \ln(\lambda) \sum_{i=1}^n x_i - \sum_{i=1}^n \ln(x_i!)$. Now we can calculate the first derivative of the Log-Likelihood w.r.t. λ , that is $\frac{dL}{d\lambda} = \frac{d}{d\lambda} \left(-n\lambda + \ln(\lambda) \sum_{i=1}^n x_i - \sum_{i=1}^n \ln(x_i!) \right) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i$. Now we can set the derivative equal to zero, $-n + \frac{1}{\lambda} \sum_{i=1}^n x_i = 0$ and rearrange to find $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i$. Therefore, the MLE is equivalent to the sample mean of the n observations in the sample, *i.e.*, $\hat{\lambda} = \bar{X}$.

MAXIMUM LIKELIHOOD ESTIMATOR FOR TWO PARAMETERS

We can also consider a probability function, $f(x; \theta_1, \theta_2)$, with two unknown parameters, θ_1 and θ_2 . Given an IID sample $X_1, X_2, \dots, X_n$, the likelihood is defined as $l(\theta_1, \theta_2; X_1, X_2, \dots, X_n) = \prod_{i=1}^n f(X_i; \theta_1, \theta_2)$, and thus, the log-likelihood is defined as $L(\theta_1, \theta_2) = \ln l(\theta_1, \theta_2) = \sum_{i=1}^n \ln f(X_i; \theta_1, \theta_2)$. The maximum likelihood estimates, $\hat{\theta}_1$ and $\hat{\theta}_2$, of the parameters θ_1 and θ_2 , thus satisfy $\frac{\partial L(\theta_1, \theta_2)}{\partial \theta_1} \Big|_{(\theta_1, \theta_2)=(\hat{\theta}_1, \hat{\theta}_2)} = 0 = 0$ and $\frac{\partial L(\theta_1, \theta_2)}{\partial \theta_2} \Big|_{(\theta_1, \theta_2)=(\hat{\theta}_1, \hat{\theta}_2)} = 0$.

Do note that there are also some conditions on the second derivatives to ensure that a maximum occurs at $(\theta_1, \theta_2) = (\hat{\theta}_1, \hat{\theta}_2)$. That is, the matrix $\begin{bmatrix} -\frac{\partial^2 L}{\partial \theta_1^2} \Big|_{(\theta_1, \theta_2)=(\hat{\theta}_1, \hat{\theta}_2)} & -\frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} \Big|_{(\theta_1, \theta_2)=(\hat{\theta}_1, \hat{\theta}_2)} \\ -\frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} \Big|_{(\theta_1, \theta_2)=(\hat{\theta}_1, \hat{\theta}_2)} & -\frac{\partial^2 L}{\partial \theta_2^2} \Big|_{(\theta_1, \theta_2)=(\hat{\theta}_1, \hat{\theta}_2)} \end{bmatrix}$ must be positive definite, *i.e.*, the top-left and bottom-right element > 0 and the determinant > 0 .

CONCLUSION

The MLE is one of the most used methods in statistics for parameter estimation for our model. Its flexibility allows it to be applied to a wide range of probability distributions and models. The British statistician, George

Box famously stated that "essentially, all models are wrong, but some are useful". The MLE can help improve models through design optimisation, ensuring an optimal solution for the parameters, making our models as useful as possible.

ALEXANDER FLANN